

Escape The Maze

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Graphics

1. Objectives

- A. Create an interactive Graphical Application
- B. Manage a Complex Scene
- C. Textures
- D. Lighting
- E. **Can you achieve all of the extra credit???**

2. Equipment

- A. A computer that can run Notepad++ and a modern web-browser

3. Lab Preparation

- A. Make sure you understand how to create and apply textures

4. Lab Instruction

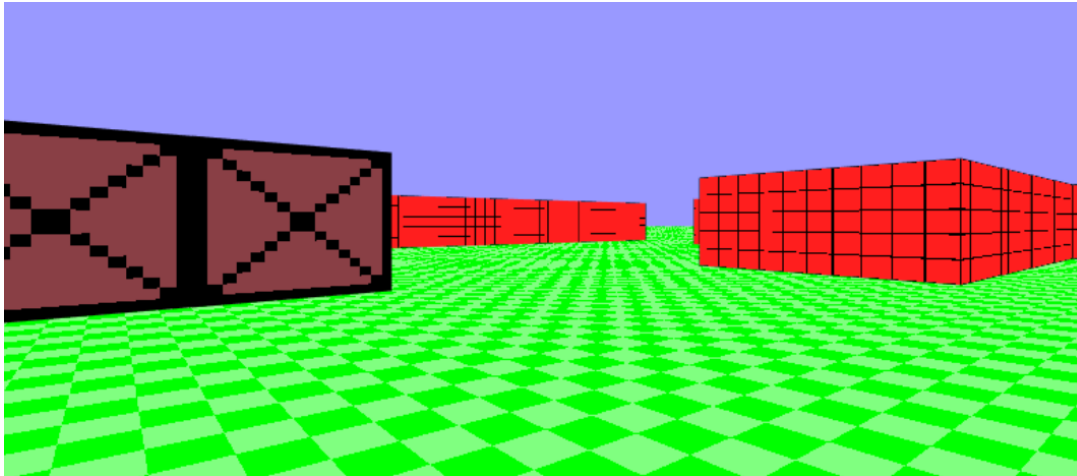
One other thing you CANNOT USE the checkered or brick texture. I want you to make your own (or host and use a picture).

- A. You will create three textures.
 - 1. One for the Floor
 - 2. One for wall type 1
 - 3. One for wall Type 2.
- B. The walls will be constructed out of 2x2x2 cubes.
- C. There must be two types of textured blocks for two different types of walls. Create a 30X30 array that will hold values for the maze.
 - 1. (for example 0 – open space, 1 wall type 1, 2 wall type 2)
- D. You will start in the maze and try to get out.
- E. You cannot go through the wall (Do not let your camera clip either.)
- F. You must have directional light and 10 point lights.
- G. If you press space bar (or any other button you wish) you will shoot a bullet out in front of you. This can be a billboarded sprite or an actual 3D object.
- H. Create three bad guys
 - 1. Bad guys will be sprite textures that are applied to a square (quad).

2. You can import your pictures into array form using this link, (warning the picture will be upside-down). <https://littlevgl.com/image-to-c-array>
3. There will be three different types of bad guys. One will patrol back and forth. One will randomly move once it hits a wall. One will randomly move and periodically shoot a bad bullet at the player. (This can be a sprite or an actual 3D object)
4. All bad guys will be bill-boarded.
5. If a bad guy collides with the camera, you must restart the level.
6. If you get hit by 5 bad bullets you must restart the level.
7. Each monster will take one damage if hit by the heroes bullets. Each monster type will have a different amount of hp. (I will let you decide). Once it reaches 0, the monster is removed.

HINT: Do not make very narrow pathways, our collision will get caught.

Example:



Grading rubric next page!

5. Lab Rubric

	Perfect	Incomplete but functional	Not functional or attempted
3 Textures	25	15	0
Cannot walk through walls	5	3	0
Directional light and 10 point lights	15	7	0
Game prints "You Win" if you reach the end	5	3	0
Monsters are bill-boarded	25	10	0
Three different monsters	10	5	0
One monster can shoot at you.	10	5	0
You can destroy monsters and they can reset you appropriately.	10	5	0
Total	/105		
Difficult (extra credit 5)*	Use three different textures for each monster to simulate animation.		
Challenging (+5 pts)*	Create a bill-boarded texture that represents an explosion, when a monster dies, spawn this shape first then destroy the monster		
Nightmare! (+25 pts)*	Finish off the project by adding a title menu, player options, three levels and sound effects. (Must have completed the other two!)... Extra bragging rights if you can include a story!		

***Note this assignment must be coded from scratch. You cannot use any external libraries!**

6. Lab Report Requirements

There is nothing for the report or peer review for this assignment.